

Doctor–Patient Portal

MEDI-CONNECT

A secure, containerized web platform digitizing end-to-end clinic operations through a custom Java MVC architecture and DevOps deployment.

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Project Introduction

Medi-Connect is a secure, role-based web application that digitizes end-to-end clinic operations — bridging the communication gap between patients, doctors, and administrators.

Domain

Digital Healthcare & Clinic Workflow Automation

Architecture

Java MVC using JSP & Servlets with strict layer separation

Deployment

Fully containerized with Docker for portable cloud delivery

Users

Three role-based dashboards — Admin, Doctor and Patient

Problem Statement

Traditional clinic management still relies on phone calls, paper registers and isolated desktop tools — creating friction for patients and overhead for medical staff.



Scheduling Conflicts

Manual booking causes double-bookings, missed slots and long wait times.



Fragmented Records

Patient data sits in silos with no unified, searchable history.



Weak Access Control

Sensitive records are exposed due to absent role-based permissions.



Poor Patient Experience

No easy way to discover specialists or track appointment status.



Limited Scalability

Hardcoded configurations make legacy systems hard to deploy or scale.



Deployment Friction

Environment inconsistencies break installs across machines and servers.

Objectives of the Project

01

Digitize Clinic Operations

Replace manual workflows with a unified, web-based portal.

03

Automate Appointment Booking

Match patients to doctors by specialization in real time.

05

Ensure Portable Deployment

Containerize the stack with Docker for reliable releases.

02

Enforce Role-Based Access

Provide isolated, secure dashboards for Admin, Doctor and Patient.

04

Centralize Patient Data

Maintain a single, consistent record of medical history.

06

Deliver a Scalable Foundation

Build on MVC patterns ready for production-grade growth.

Proposed System

Medi-Connect offers a containerized Java web application built on the MVC pattern, delivering a secure, role-aware portal that unifies clinic operations under a single interface.

Core Capabilities

- Three dedicated dashboards — Admin, Doctor, Patient
- Automated doctor matching based on medical specialization
- Real-time appointment scheduling with live status tracking
- Secure session management protecting sensitive health data
- Centralized MySQL store for patients, doctors and bookings
- Docker-based deployment with automated DB initialization

Secure

RBAC + Sessions

Scalable

MVC Architecture

Portable

Dockerized Stack

Smart

Specialization Match

Technologies Used

Frontend

JSP • HTML5 • CSS3 • JavaScript

Backend

Core Java • Servlets • JDBC

Database

MySQL Relational Database

Server

Apache Tomcat Container

DevOps

Docker • docker-compose

Pattern

Model–View–Controller (MVC)

System Architecture & Workflow

Presentation Layer

JSP Pages • Browser UI • CSS / JS

Controller Layer

Java Servlets • Request Routing • Session Mgmt

Model / Business Layer

DAO Classes • Business Logic • Validation

Data Layer

MySQL Database • JDBC Connectivity

Request Lifecycle

1

Patient logs in via secure session

2

Browses doctors by specialization

3

Servlet routes booking request

4

DAO validates & persists to MySQL

5

Dashboard reflects live status

Key Features & Modules

Admin Module

- ▶ Manage doctors (CRUD)
- ▶ Oversee specializations
- ▶ Monitor platform metrics

Doctor Module

- ▶ Personalized dashboard
- ▶ View patient appointments
- ▶ Update appointment status

Patient Module

- ▶ Search doctors by specialty
- ▶ Book real-time appointments
- ▶ Track booking lifecycle

Security & Sessions

- ▶ Role-Based Access Control
- ▶ Encrypted credentials
- ▶ Server-side session checks

Specialist Matcher

- ▶ Auto-mapping by category
- ▶ Filtered doctor listings
- ▶ Reduced search friction

Deployment Module

- ▶ Dockerfile + Compose
- ▶ Auto DB initialization
- ▶ One-command launch

Database & Implementation Overview

users

user_id (PK)
email, password
role (admin/doctor/patient)

specialist

spec_id (PK)
specialization_name
description

doctor

doctor_id (PK)
user_id (FK)
spec_id (FK)

appointment

appt_id (PK)
patient_id, doctor_id (FK)
date, time, status

Implementation Highlights

- JDBC connection pool via singleton DB utility
- Servlet controllers for login, booking, admin ops
- DAO layer isolates SQL from business logic
- JSP views with shared header/footer fragments
- Validation on both client and server side
- docker-compose orchestrates app + MySQL services
- Schema auto-loaded on first container boot

Results & Outcomes

3

Role-Based Dashboards

4

Core Database Modules

1-Cmd

Docker Deployment

100%

MVC Layer Separation

Validated Outcomes

Functional Testing

All booking, login, and admin flows passed unit and integration tests.

Deployment Reliability

Docker build verified across multiple environments without configuration drift.

Security Validation

Session hijacking and unauthorized access attempts were blocked by RBAC.

Usability

Clean, responsive UI confirmed positive feedback on dashboards and forms.

Conclusion & Future Scope

Conclusion

Medi-Connect successfully delivers a secure, role-aware web portal that replaces error-prone manual workflows with an automated, containerized platform. The MVC foundation, combined with Docker-based deployment, results in a maintainable, production-ready system that meaningfully improves how patients, doctors and administrators interact within a modern clinic.

Future Scope

- ▶ Integrated tele-consultation with video calling
- ▶ Online prescription generation and e-pharmacy linkage
- ▶ Patient health analytics and visit history dashboards
- ▶ Mobile applications for Android and iOS
- ▶ AI-assisted symptom checker and doctor recommendation
- ▶ Integration with payment gateways and insurance APIs